

OMKARNATH THAKUR

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Education

University of Maryland, College Park

Master of Science in Data Science

Jan 2024 – Dec 2025

College Park, Maryland

Technical Skills

Languages: Python, SQL, R, Java

Backend/Frontend: FastAPI, Flask, Node.js, Express.js, Next.js, HTML, JavaScript/TypeScript, MongoDB

Pipelines: Machine Learning, Deep Learning, Generative AI, Natural Language Processing, RAG, Agents, Prompt Engineering

Frameworks: PyTorch, Pandas, NLTK, TF-IDF, spaCy, NumPy, Scikit-learn, TensorFlow, KerasTuner, LangChain

AI Tools: LangGraph, CrewAI, Hugging Face, AgentSDK

Deployment & Monitoring: MCP, Docker, Kubernetes, Grafana, GitHub Actions, Airflow, MLflow, Terraform, AWS, Azure, GCP

Training & Evaluation: Fine-tuning, PEFT, QLoRA, Knowledge Distillation, Mixture of Experts, Cost Optimization, vLLM

Experience

Machine Learning Engineer

Feb 2026 – Present

Community Dreams Foundation

- Built a **GPT-4** multi-agent system with **CrewAI** and **Pydantic** to rank RSS-feed deals, reducing manual effort by **90%** through multi-source deal scoring, ranking, and aggregation for scalable market intelligence.
- Engineered a custom **RAG** pipeline with **ChromaDB** and 3D embedding visualization, improving hybrid product-similarity search accuracy by **20%**.
- Developed an **LLM + Random Forest** ensemble, deployed it on **Modal**, and enabled real-time monitoring through **Gradio**, lowering price-prediction error by **15%**.
- Designed and deployed a multi-agent financial planning system on **AWS**, orchestrating Planner, Reporter, Chart Maker, Retirement Specialist, and Researcher agents to generate portfolio analyses with shared **S3** vector knowledge.
- Improved agent reliability with **Pydantic** structured outputs for InstrumentTagger, validated Chart Maker JSON, and added input sanitization, prompt-injection checks, and retry logic to reduce invalid outputs.
- Evaluated **AWS Bedrock AgentCore** as a managed agent runtime against a **Terraform**-provisioned **Lambda + API Gateway + Aurora + SageMaker** architecture to assess deployment, tooling, and observability tradeoffs.

Artificial Intelligence Intern

Jun 2025 – Aug 2025

The Builder Market

Los Angeles, CA

- Developed a production-grade **Agentic RAG** assistant using **MERN**, **PostgreSQL**, **LangChain**, and **OpenAI APIs** to answer construction workflow queries, improving response grounding through hierarchical chunking and reranking.
- Evaluated retrieval performance using **MRR**, **nDCG**, and **LLM-as-a-Judge**, tuning chunking and ranking strategies to improve answer relevance and overall retrieval quality.
- Engineered a large-scale ML pipeline processing **400K+** product descriptions across **8** categories using **Python**, **Hugging Face**, and **OpenAI APIs**, benchmarking **12** traditional ML and LLM-based models.
- Fine-tuned **GPT-4o-mini** on **JSONL** datasets with **Weights & Biases** tracking, achieving **70–80%** accuracy, **15–25%** error, and **RMSLE 0.3–0.5**, outperforming baselines by **40–50%**.

Data Scientist

Jun 2020 – Jul 2023

Kamdhenu Robotics

Bengaluru, India

- Engineered an end-to-end call summarization solution by fine-tuning a **BERT** model and automating deployment to **AWS SageMaker** through Dockerized **GitHub Actions** CI/CD, saving analysts **30 minutes** of reporting time per call.
- Optimized experimentation efficiency by developing a **K-Means** clustering pipeline for time-series peak detection and applying **ARIMA** and regression models for forecasting, cutting redundant runs and saving approximately **\$50K annually**.
- Developed a transformer-based **Named Entity Recognition (NER)** pipeline with **80%** accuracy to extract vendors, products, and markets from call transcripts, improving insight generation from unstructured business text.
- Orchestrated a multi-modal AI pipeline with **FastAPI** to process PDFs, images, CSVs, PPTs, and raw text, converting heterogeneous inputs into usable text and audio outputs through a unified workflow.
- Deployed a **RAG** pipeline with **LoRA**-tuned **Mistral-7B**, enabling grounded responses and improved answer quality across structured and unstructured data sources.

Projects

OpenTSLM Paper Implementation | Python, PyTorch, Hugging Face, CUDA, Llama, Gemma

2025

- Reproduced **OpenTSLM** by extending pretrained **Llama/Gemma** models for reasoning over multivariate time-series data.
- Benchmarked training/evaluation across HAR, SleepEDF, ECGQA, M4 using curriculum and multimodal inference workflows.
- Validated pretrained checkpoints with **CUDA**-based inference and structured benchmarking to improve experimentation efficiency.

Curious RL Paper Implementation | Python, PyTorch, Reinforcement Learning, A2C, VizDoom

2025

- Reproduced exploration paper by integrating an **Intrinsic Curiosity Module** with the **A2C** reinforcement algorithm in Python.
- Trained/evaluated agents on **VizDoom** to replicate exploration behavior and benchmark performance against vanilla **A2C**.
- Analyzed training stability with **TensorBoard** monitored and regulated, identified local-optima failure modes and agent learning.